under Captain Ryurik Ketov, was detected and pursued by ASW aircraft, but avoided having to surface in the presence of the US navy.

One of the more remarkable suggestions about the operation is that senior officials in the Soviet Ministry of Defence did not realise the submarines were powered by diesel-electric engines and not nuclear reactors. The First Deputy Minister of Defence, Marshal Andrei Grechko, is reported to have believed the submarines were nuclear-powered and had therefore no need to surface.[37] Any captain who obeyed an instruction to surface would been disobeying direct orders (as well as shaming the navy). Indeed, several senior naval officers believed the commanders should have fired their weapons rather than surface their boats.[38] If the First Deputy Minister of Defence was unaware of this crucial operational factor, what did Khrushchev and his colleagues in the Presidium know or understand?

## ‘Sink them all’

Of greatest importance was command and control of the nuclear torpedoes. Soviet naval procedures were different from those for land- or air-based systems, where warheads and bombs were usually stored far away from the delivery systems. Movement of warheads was strictly controlled, involving a rigorous system of authorisation (and the use of specially dedicated KGB troops). On the submarines, the nuclear torpedoes were stored in the forward torpedo room, along with conventionally armed weapons. Preparing the T-5 for battle was the responsibility of a special weapons security officer, on whom the arming sequence therefore depended.

The B-130 was one of the submarines targeted by US ASW forces using the SSIPs. According to Peter Hutchausen’s 2002 account, when Captain Shumkov’s boat was subjected to continuing bombardment, he ordered the loading of his nuclear torpedo. The special weapon security officer told him that it could not be armed without specific instructions from the Special Weapons Directorate of the Main Navy Staff. ‘Look, just do as you’re told, and I’ll handle the permission,’ Shumkov replied (according to Shumkov).[39] Shortly afterwards, the special weapons security officer passed out. Captain Shumkov did not intend to fire the weapon, so he explained to his deputy